

2024 Calculus II Additional Exercise (13)

1. Exercise 2, 4, 5, in page 53 from Calculus Supplement Homework.
2. Exercise 14 (where f has continuous derivatives on $(-\infty, \infty)$ and $f(0) = 0$) in page 53 from Calculus Supplement Homework.
3. Exercise 15 in page 53 from Calculus Supplement Homework.
4. Exercise 22 in page 54 from Calculus Supplement Homework.
5. Exercise 43 in page 56 from Calculus Supplement Homework.(Hint: Using Green's Theorem with a hole)

Q1 P53 Ex 2

2. Assume L is the ellipse $4x^2 + y^2 = 8x$ in the counterclockwise direction, then $\oint_L e^{y^2} dx + x dy = (\quad)$.

Sol.

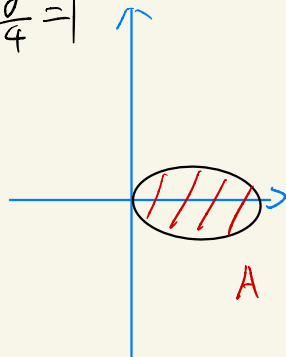
$$\oint_L e^{y^2} dx + x dy$$

$$(\text{Green's Thm}) = \iint_A \left(\frac{\partial x}{\partial x} - \frac{\partial e^{y^2}}{\partial y} \right) dx dy$$

$$= \iint_A 1 - 2ye^{y^2} dx dy$$

$$= 2\pi$$

$$(x-1)^2 + \frac{y^2}{4} = 1$$



关于 y 的奇函数在关于 x 轴对称
区间上积分为 0.

PS3 Ex 4, 5

4. Assume $\frac{(x+ay)dy-ydx}{(x+y)^2}$ is exact, then $a = (\quad)$.

5. Let L be the counterclockwise circle $x^2 + (y-1)^2 = 4$, then $\oint_L \frac{x dy - y dx}{x^2 + (y-1)^2} = (\quad)$.

Sol to Ex 4: $a = 0$ $d\left(\frac{y-x}{2(x+y)}\right) = \frac{xdy-ydx}{(x+y)^2}$

Reason: Exact 1-form must be closed 1-form

$$\text{Let } \begin{cases} f(x,y) = \frac{x+ay}{(x+y)^2} \\ g(x,y) = \frac{-y}{(x+y)^2} \end{cases}$$

$$\therefore 0 = d(f(x,y)dy + g(x,y)dx) = \left(\frac{\partial f}{\partial x} - \frac{\partial g}{\partial y}\right) dx \wedge dy$$

$$\Rightarrow \frac{\partial f}{\partial x} = \frac{\partial g}{\partial y}$$

$$\frac{(x+y)^2 - 2(x+y)(x+ay)}{(x+y)^4} = \frac{-(x+y)^2 + 2y(x+y)}{(x+y)^4}$$

$$x+y-2(x+ay) = -(x+y) + 2y$$

$$-2ay = 0$$

$$a = 0$$

Sol to Ex 5: 参数化 $\begin{cases} x = 2\cos t \\ y = 2\sin t + 1 \end{cases} \Rightarrow \begin{cases} x'(t) = -2\sin t \\ y'(t) = 2\cos t \end{cases} \quad (0 \leq t \leq 2\pi)$ □

$$\therefore \oint_L \frac{xdy-ydx}{x^2+(y-1)^2} = \frac{1}{4} \int_0^{2\pi} 4\cos^2 t + (2\sin t + 1) \cdot 2\sin t dt$$

$$= 2\pi + \int_0^{2\pi} 2\sin t dt$$

$$= 2\pi$$

(因为 1-form 在 (0,0) 处无定义, 所以无法对 Region $\{x^2+(y-1)^2 \leq 4\}$ 用 Green 公式)

□

Q2 P53 Ex 14

14. Assume that the integral $\int_C (f(x) - e^x) \sin y \, dx - f(x) \cos y \, dy$ is independent with respect to the path. If f has continuous derivatives and $f(0) = 0$, find $f(x)$.

pt: Above 1-form is defined on $\mathbb{R}^2$, which is connected, simply connected.

Green formula $0 = \oint_C (f(x) - e^x) \sin y \, dx - f(x) \cos y \, dy$

$$= \iint_D -f'(x) \cos y - (f(x) - e^x) \cos y \, dx \, dy$$

which holds for all region $D \subseteq \mathbb{R}^2$.

$$\therefore -f'(x) \cos y - (f(x) - e^x) \cos y = 0$$

$$f'(x) + f(x) = e^x$$

$$(f'(x) + f(x))e^x = e^{2x}$$

$$d(f(x) \cdot e^x) = e^{2x} dx$$

$$f(x)e^x = \frac{1}{2} e^{2x} + C$$

$$\therefore f(0) = 0 \Rightarrow C = -\frac{1}{2} \quad f(x) = \frac{1}{2} (e^x - e^{-x})$$

□

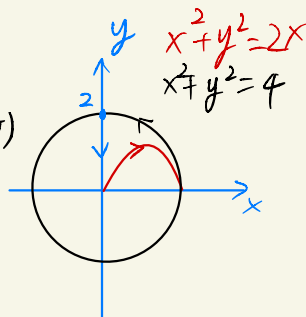
Q3 PS3 Ex15

15. Compute $\int_C \vec{F} \cdot d\vec{r}$, where $\vec{F}(x, y) = (3x^2y, x^3 + x - 2y)$, and the curve C starts at $(0, 0)$, moves to $(2, 0)$ along $x^2 + y^2 = 2x$ in the first quadrant, then travels to $(0, 2)$ along $x^2 + y^2 = 4$ in the first quadrant.

Sol: 法一: 分两段求:

$$C_1: (x-1)^2 + y^2 = 1 \quad \text{参数化} \begin{cases} x = (\cos(\pi-t) + 1) = -\cos t + 1 \\ y = \sin(\pi-t) = \sin t \end{cases} \quad (0 \leq t \leq \pi)$$

$$C_2: x^2 + y^2 = 4 \quad \text{参数化} \begin{cases} x = 2\cos\theta \\ y = 2\sin\theta \end{cases} \quad (0 \leq \theta \leq \frac{\pi}{2})$$



$$\Rightarrow \begin{cases} x'(t) = \sin t \\ y'(t) = \cos t \end{cases} \quad \begin{cases} x'(\theta) = -2\sin\theta \\ y'(\theta) = 2\cos\theta \end{cases}$$

$$\begin{aligned} \text{Then } \int_{C_1} \vec{F} d\vec{r} &= \int_{C_1} 3x^2y x' + (x^3 + x - 2y) y' dt \\ &= \int_0^\pi 3(-\cos t + 1)^2 \sin^2 t + ((-\cos t + 1)^3 + (-\cos t + 1) - 2\sin t) \cos t dt \\ &= \text{不好算} \end{aligned}$$

$$\begin{aligned} \int_{C_2} \vec{F} d\vec{r} &= \int_{C_2} 3x^2y x' + (x^3 + x - 2y) y' d\theta \\ &= \int_0^{\frac{\pi}{2}} 3 \cdot (2\cos\theta)^2 \cdot 2\sin\theta \cdot (-2\sin\theta) + ((2\cos\theta)^3 + 2\cos\theta - 4\sin\theta) \cdot 2\cos\theta d\theta \\ &= \text{不好算} \end{aligned}$$

$$\int_C \vec{F} d\vec{r} = \int_{C_1} \vec{F} d\vec{r} + \int_{C_2} \vec{F} d\vec{r}.$$

法二: Green's Theorem;

$$C_3: \begin{cases} x=0 \\ y=2-t \end{cases} \quad (0 \leq t \leq 2) \Rightarrow \begin{cases} x'=0 \\ y'=-1 \end{cases}$$

$$\begin{aligned} \therefore \int_{C_3} \vec{F} d\vec{r} &= \int_0^2 3x^2 y x' + (x^3 + x - 2y) y' dt \\ &= \int_0^2 -2(2-t) \cdot (-1) dt \\ &= 8 - t^2 \Big|_0^2 \\ &= 4 \end{aligned}$$

Use Green's Theorem:

$$\begin{aligned} \int_C \vec{F} d\vec{r} + \int_{C_3} \vec{F} d\vec{r} &= \oint_{\tilde{C}} 3x^2 y dx + (x^3 + x - 2y) dy \\ &= \iint_D \frac{\partial (x^3 + x - 2y)}{\partial x} - \frac{\partial 3x^2 y}{\partial y} dx dy \\ &= \iint_D dx dy \\ &= \frac{1}{4} \pi \cdot 2^2 - \frac{1}{2} \pi \cdot 1^2 \\ &= \frac{\pi}{2} \end{aligned}$$

$$\therefore \int_C \vec{F} d\vec{r} = \frac{\pi}{2} - 4$$

□

Q4 P54 Ex22

22. Consider the following line integral:

$$\int_{(1,1,1)}^{(1,3,\pi)} e^x \ln y \, dx + \left(\frac{e^x}{y} + \sin z \right) dy + y \cos z \, dz.$$

Show that the differential form in the integral is exact and evaluate the integral.

Sol: $d(e^x \ln y + y \sin z) = e^x \ln y \, dx + \left(\frac{e^x}{y} + \sin z \right) dy + y \cos z \, dz$

is an exact 1-form.

$$\begin{aligned} \therefore \int_{(1,1,1)}^{(1,3,\pi)} d(e^x \ln y + y \sin z) &= e^x \ln y + y \sin z \Big|_{(1,1,1)}^{(1,3,\pi)} \\ &= e \ln 3 - \sin 1 \end{aligned}$$

D

Q5 P56 Ex 43

43. Let C be the circle $x^2 + y^2 = 9$ in the counterclockwise direction. Compute

$$\oint_C (3y - e^{\sin x}) dx + (7x + \sqrt{y^4 + 1}) dy.$$

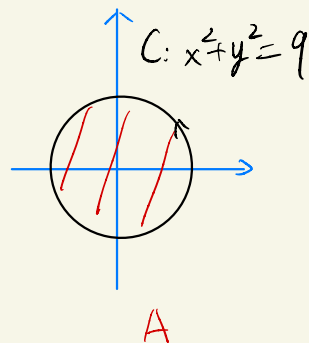
Sol. By Green's Thm,

$$\text{原式} = \iint_A \frac{\partial (7x + \sqrt{y^4 + 1})}{\partial x} - \frac{\partial (3y - e^{\sin x})}{\partial y} dx dy$$

$$= \iint_A 4 dx dy$$

$$= 4 \cdot \pi \cdot 3^2$$

$$= 36\pi$$



Q

超纲内容: 用微分形式的语言叙述梯度、旋度、散度、Green's Thm, Stokes' Thm, Divergence Thm.

Def: 0-form: $\Omega^0(\mathbb{R}^3) = \{f \in C^\infty(\mathbb{R}^3)\}$

1-form: $\Omega^1(\mathbb{R}^3) = \{w = f dx + g dy + h dz \mid f, g, h \in C^\infty(\mathbb{R}^3)\}$

2-form: $\Omega^2(\mathbb{R}^3) = \{w = f dx \wedge dy + g dy \wedge dz + h dz \wedge dx \mid f, g, h \in C^\infty(\mathbb{R}^3)\}$

$$\left. \begin{array}{l} \text{with } dx \wedge dy = -dy \wedge dx \quad dx \wedge dx = -dx \wedge dx \\ dy \wedge dz = -dz \wedge dy \quad dy \wedge dy = -dy \wedge dy \\ dz \wedge dx = -dx \wedge dz \quad dz \wedge dz = -dz \wedge dz \end{array} \right\}$$

3-form: $\Omega^3(\mathbb{R}^3) = \{w = f dx \wedge dy \wedge dz \mid f \in C^\infty(\mathbb{R}^3) \text{ with } dx, dy, dz\}$
 两两交换一次顺序乘一个负号

Def (外微分算子 d)

$$\Omega^0(\mathbb{R}^3) \xrightarrow{d} \Omega^1(\mathbb{R}^3) \xrightarrow{d} \Omega^2(\mathbb{R}^3) \xrightarrow{d} \Omega^3(\mathbb{R}^3)$$

$$f \mapsto df = f'_x dx + f'_y dy + f'_z dz$$

$$P dx + Q dy + R dz \mapsto dP \wedge dx + dQ \wedge dy + dR \wedge dz \\ \underline{(Q'_x - P'_y) dx \wedge dy + (R'_y - Q'_z) dy \wedge dz + (P'_z - R'_x) dz \wedge dx}$$

$$P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy \mapsto dP \wedge dy \wedge dz + dQ \wedge dz \wedge dx + dR \wedge dx \wedge dy \\ \underline{(P'_x + Q'_y + R'_z) dx \wedge dy \wedge dz.}$$

Rmk: (1) 可以看出 $d \circ d = 0$ 总是成立. 那么根据下页的对应关系:

$$\nabla \times (F) = 0, \quad \nabla \cdot (\nabla \times F) = 0 \text{ 总成立.}$$

(2) 事实上, 用微分形式的语言叙述 Stokes' Thm 和 divergence Thm 可以统一成 $\int_{\partial R} w = \int_R dw$, 其中 w 分别取 1-form 和 2-form 的情况

对应: f : 可微函数

∂S : 曲面 S 的边界

$\vec{F} = P\vec{i} + Q\vec{j} + R\vec{k}$: 向量场

∂D : 三维物体的边界

向量场

微分形式

梯度

$$\nabla f = f'_x \vec{i} + f'_y \vec{j} + f'_z \vec{k}$$

$$df = f'_x dx + f'_y dy + f'_z dz \quad 1\text{-form}$$

旋度

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

$$= (R'_y - Q'_z) \vec{i} + (P'_z - R'_x) \vec{j} + (Q'_x - P'_y) \vec{k}$$

$$\begin{aligned} d(Pdx + Qdy + Rdz) \\ = (R'_y - Q'_z) dy \wedge dz + (P'_z - R'_x) dz \wedge dx \\ + (Q'_x - P'_y) dx \wedge dy \quad 2\text{-form} \end{aligned}$$

散度

$$\nabla \cdot \vec{F} = P'_x + Q'_y + R'_z$$

$$\begin{aligned} d(Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy) \\ = (P'_x + Q'_y + R'_z) dx \wedge dy \wedge dz \quad 3\text{-form} \end{aligned}$$

Stokes' Thm

$$\oint_{\partial S} \vec{F} \cdot \vec{T} ds = \iint_S (\nabla \times \vec{F}) \cdot \vec{n} d\sigma$$

$$\begin{aligned} \oint_{\partial S} Pdx + Qdy + Rdz \\ \stackrel{\text{Stokes' Thm}}{=} \iint_S d(Pdx + Qdy + Rdz) \\ = \iint_S (R'_y - Q'_z) dy \wedge dz + (P'_z - R'_x) dz \wedge dx \\ + (Q'_x - P'_y) dx \wedge dy \end{aligned}$$

Divergence Thm

$$\oint_{\partial D} \vec{F} \cdot \vec{n} d\sigma = \iiint_D \nabla \cdot \vec{F} dV$$

$$\begin{aligned} \oint_{\partial D} Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy \\ \stackrel{\text{Divergence Thm}}{=} \iiint_D d(Pdy \wedge dz + Qdz \wedge dx + Rdx \wedge dy) \\ = \iiint_D (P'_x + Q'_y + R'_z) dx \wedge dy \wedge dz \end{aligned}$$